A Fully Parametric Sequential Cure Model with Treatment-Dependent Copula Association

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Abstract

Sequential survival data with a cure fraction combine induced censoring of later gap times with structural absence of later states for cured subjects. We propose a fully parametric model for two sequential gap times that combines a mixture cure model for the first gap with a Clayton copula whose association parameter may vary with treatment or other covariates. Marginal, cure, and association parameters are estimated jointly by maximum likelihood with sandwich inference. Simulations show small bias and generally near-nominal coverage when dependence is correctly modeled. A focused comparison with a constant-dependence copula shows that incorrectly imposing homogeneous association can introduce persistent bias in marginal treatment-effect estimates, with increasingly poor confidence-interval coverage at larger sample sizes. In the colon cancer application, treatment effects differ across the two gap-time margins, but there is little evidence that treatment modifies their residual association.

1 Introduction

Sequential event-time data arise when subjects move through ordered disease states, such as recurrence followed by post-recurrence death. If T_1 is the first gap time and T_2 the subsequent gap, then T_2 is observable only after the first event. Even under independent administrative censoring, the follow-up available for T_2 depends on the realized T_1, producing induced dependent censoring (Lawless & Yilmaz, 2011; Lin et al., 1999). This setting is related to, but distinct from, semi-competing risks, where a terminal event may occur before and preclude a non-terminal event (Fine et al., 2001); the present work focuses on a strictly sequential gap-time representation.

A further complication is that some subjects may never experience the first event. Mixture cure models separate susceptibility from the event-time distribution among susceptible subjects (Farewell, 1982). For sequential gap times, Lawless and Yilmaz (2011) developed a semiparametric copula framework that addresses the cure fraction and induced censoring of the second gap.

Copula models separate marginal event-time behavior from residual association (Shih & Louis, 1995). However, treating association as constant across subjects can be restrictive. Covariate-dependent copulas have been studied by Acar et al. (2011) and, for censored clustered event times, by Geerdens et al. (2018). Treatment-dependent association has also been considered for correlated survival endpoints (Emura et al., 2021). These developments motivate allowing treatment to modify association in the sequential cure setting rather than treating dependence as a fixed nuisance quantity. An additional concern is that imposing a common association parameter when dependence is heterogeneous may affect estimation of the marginal treatment effects through the joint likelihood.

We therefore combine a sequential cure likelihood with regression on the Clayton association parameter. Both susceptible gap-time margins are modeled parametrically through accelerated failure time distributions, while the copula parameter varies with treatment or other covariates. All components are estimated jointly by maximum likelihood. We evaluate finite-sample estimation and Wald inference over varying sample sizes, censoring levels, and association heterogeneity. We also examine the consequences of restricting the dependence to be constant, with particular attention to whether treatment-effect bias and confidence-interval undercoverage diminish as sample size increases. The method is illustrated with the colon cancer trial data from the R `survival` package.

2 Methodology

For subject $i = 1, \ldots, n$, let T_{1i} denote the time from study entry to the first event and T_{2i} the elapsed time from the first event to the second event. Let C_i denote an independent end-of-follow-up time measured from study entry, and let $I(\cdot)$ denote the indicator function. The observed first gap time and indicator are $Y_{1i} = \min(T_{1i}, C_i)$ and $\delta_{1i} = I(T_{1i} \leq C_i)$. When $\delta_{1i} = 1$, the remaining follow-up time is $C_i - T_{1i}$, and the second gap contributes $Y_{2i} = \min\{T_{2i}, C_i - T_{1i}\}$ and $\delta_{2i} = I\{T_{2i} \leq C_i - T_{1i}\}$. Let $\boldsymbol{x}_{1i}$ and $\boldsymbol{x}_{2i}$ denote the covariate vectors for the first and second gap-time margins, respectively. We allow a fraction of subjects never to experience the first event. Let $p = P(T_{1i} < \infty)$ be the susceptible proportion and $1 - p$ the cure proportion. Conditional on susceptibility, define

$$F_1(t_1 \mid \boldsymbol{x}_{1i}) = P(T_{1i} \leq t_1 \mid T_{1i} < \infty, \boldsymbol{x}_{1i}), \tag{1}$$

$$F_2(t_2 \mid \boldsymbol{x}_{2i}) = P(T_{2i} \leq t_2 \mid T_{1i} < \infty, \boldsymbol{x}_{2i}). \tag{2}$$

The unconditional distribution of the first gap is therefore

$$P(T_{1i} \leq t_1 \mid \boldsymbol{x}_{1i}) = pF_1(t_1 \mid \boldsymbol{x}_{1i}), \tag{3}$$

and its survivor function for finite t_1 is $1 - pF_1(t_1 \mid \boldsymbol{x}_{1i})$. Conditional on $T_{1i} < \infty$, we join the two marginal distributions with a Clayton copula,

$$C_{\phi_i}(u, v) = \left(u^{-\phi_i} + v^{-\phi_i} - 1\right)^{-1/\phi_i}, \qquad \phi_i > 0, \tag{4}$$

where

$$u = F_1(t_1 \mid \boldsymbol{x}_{1i}), \qquad v = F_2(t_2 \mid \boldsymbol{x}_{2i}). \tag{5}$$

The joint conditional distribution is

$$P(T_{1i} \le t_1, T_{2i} \le t_2 \mid T_{1i} < \infty, \boldsymbol{x}_{1i}, \boldsymbol{x}_{2i}) = C_{\phi_i}(u, v). \tag{6}$$

To permit association to vary across observed characteristics, the Clayton parameter is linked to a dependence-regression vector $\boldsymbol{z}_i$ through

$$\phi_i = g(\boldsymbol{z}_i^\top \boldsymbol{\gamma}). \tag{7}$$

Here $g(\cdot)$ is a positive link function; throughout this paper we use $g(a) = \exp(a)$. This is the same transformation used in treatment-dependent conditional Clayton models for correlated survival endpoints (Emura et al., 2021). For the Clayton family, the corresponding Kendall's tau is

$$\tau_i = \frac{\phi_i}{\phi_i + 2}. \tag{8}$$

Let $f_1(\cdot \mid \boldsymbol{x}_{1i})$ and $f_2(\cdot \mid \boldsymbol{x}_{2i})$ denote the susceptible marginal densities, and define

$$u_i = F_1(Y_{1i} \mid \boldsymbol{x}_{1i}), \qquad v_i = F_2(Y_{2i} \mid \boldsymbol{x}_{2i}), \tag{9}$$

and write $f_{1i} = f_1(Y_{1i} \mid \boldsymbol{x}_{1i})$ and $f_{2i} = f_2(Y_{2i} \mid \boldsymbol{x}_{2i})$ for subjects entering the second state. For the copula terms, write

$$C_1(u, v, \phi) = \frac{\partial C_\phi(u, v)}{\partial u}, \qquad c_\phi(u, v) = \frac{\partial^2 C_\phi(u, v)}{\partial u \, \partial v}. \tag{10}$$

The full likelihood function including conditional dependence and the susceptible proportion under a Clayton copula association is then given by

$$\begin{aligned} L(\boldsymbol{\Psi}) = \prod_{i=1}^{n} & p^{\delta_{1i}} \left[f_{1i} f_{2i} c_{\phi_i}(u_i, v_i) \right]^{\delta_{1i}\delta_{2i}} \\ & \times \left[f_{1i} \{1 - C_1(u_i, v_i, \phi_i)\} \right]^{\delta_{1i}(1-\delta_{2i})} \left[1 - p u_i \right]^{1-\delta_{1i}}, \end{aligned} \tag{11}$$

where $\boldsymbol{\Psi}$ collects p, the two sets of marginal parameters, and $\boldsymbol{\gamma}$. Equation (11) is the fully parametric sequential cure likelihood, with the key extension that ϕ_i varies with covariates. Estimation is performed using an unconstrained optimization-scale parameter vector $\boldsymbol{\eta}$ corresponding to $\boldsymbol{\Psi}$, with logit and log transformations for constrained parameters. Let H be the Hessian of the total negative log-likelihood with respect to $\boldsymbol{\eta}$ evaluated at the estimate, and let $\boldsymbol{s}_i$ denote the corresponding individual score vector. Define

$$B = \sum_{i=1}^{n} \boldsymbol{s}_i \boldsymbol{s}_i^\top. \tag{12}$$

A sandwich covariance estimator on the optimization scale is

$$\widehat{V}_{\boldsymbol{\eta}} = H^{-1} B H^{-1}. \tag{13}$$

3 Monte Carlo Simulation

3.1 Main Monte Carlo Experiment

A Monte Carlo study was conducted to evaluate finite-sample estimation and inference for the cure, marginal, and treatment-dependent association parameters. Each setting used $R = 1000$ replications. Sample size was varied over $n = 250$ and $n = 500$. Subjects were assigned to the reference group, Treatment 1, and Treatment 2 with probabilities 0.40, 0.30, and 0.30, respectively. Let $G_i \in \{0, 1, 2\}$, with $G_i = 0$ as the reference group, and define treatment indicators

$$D_{1i} = I(G_i = 1), \qquad D_{2i} = I(G_i = 2). \tag{14}$$

The susceptible gap times were generated from

$$\log T_{1i} = a_0 + a_1 D_{1i} + a_2 D_{2i} + \varepsilon_{1i}, \tag{15}$$

$$\log T_{2i} = b_0 + b_1 D_{1i} + b_2 D_{2i} + \varepsilon_{2i}, \tag{16}$$

where $(a_0, a_1, a_2) = (0, 2, 1)$, $(b_0, b_1, b_2) = (0, 2, 1)$, both scale parameters were one, and the marginal errors followed standard Gumbel distributions. The susceptible probability was fixed at $p = 0.70$.

Dependence was generated from the Clayton copula with

$$\phi_i = \exp(\gamma_0 + \gamma_1 D_{1i} + \gamma_2 D_{2i}). \tag{17}$$

Three association settings were considered on the Kendall's tau scale, as shown in Table 1. For each group, $\phi = 2\tau/(1-\tau)$, with the corresponding $\boldsymbol{\gamma}$ determined by the log-link model.

Table 1: Association settings used in the Monte Carlo study.

Setting	Reference	Treatment 1	Treatment 2
Homogeneous	0.40	0.40	0.40
Moderate heterogeneity	0.40	0.45	0.60
Strong heterogeneity	0.40	0.55	0.70

Administrative censoring was generated as $C_i \sim \text{Uniform}(0, u_c)$. The values of u_c were calibrated empirically to obtain 20% and 40% censoring among susceptible subjects. Table 2 gives the cross-setting summary. Convergence was 99.9–100% in all settings. Under homogeneous dependence, the joint Wald test of $H_0 : \gamma_1 = \gamma_2 = 0$ remained close to the nominal 5% level. Under heterogeneous dependence, rejection increased with sample size and association strength and decreased under heavier censoring. Across all settings, the maximum absolute bias in treatment-specific Kendall's tau remained below 0.02.

Table 2: Summary of Monte Carlo performance across the 12 principal settings.

Association setting	n	Censoring (%)	Conv. (%)	Max $\lvert\text{Bias}(\widehat{\gamma})\rvert$	Max $\lvert\text{Bias}(\widehat{\tau})\rvert$	Coverage for γ	Wald rejection (%)
Homogeneous	250	20	100.0	0.017	0.002	0.944–0.968	4.0
Homogeneous	500	20	100.0	0.025	0.003	0.955–0.965	5.2
Homogeneous	250	40	100.0	0.020	0.020	0.927–0.958	7.1
Homogeneous	500	40	100.0	0.021	0.007	0.955–0.958	4.9
Moderate heterogeneity	250	20	100.0	0.035	0.003	0.935–0.955	55.1
Moderate heterogeneity	500	20	100.0	0.014	0.002	0.953–0.955	86.7
Moderate heterogeneity	250	40	99.9	0.091	0.019	0.923–0.950	49.7
Moderate heterogeneity	500	40	100.0	0.049	0.013	0.945–0.949	77.8
Strong heterogeneity	250	20	100.0	0.043	0.003	0.952–0.956	92.5
Strong heterogeneity	500	20	100.0	0.012	0.001	0.951–0.952	99.6
Strong heterogeneity	250	40	99.9	0.106	0.019	0.935–0.955	88.4
Strong heterogeneity	500	40	100.0	0.060	0.009	0.925–0.947	99.6

Table 3 reports the moderate-heterogeneity setting with $n = 250$ and 40% censoring. The cure and marginal location parameters remained nearly unbiased, with empirical standard deviations generally close to the mean sandwich standard errors. The main deterioration occurred for the scale parameters, whose 95% coverages were 0.899 and 0.890, and for γ_1, where the bias was 0.091 and the mean SE of 0.527 was smaller than the empirical SD of 0.638. In contrast, γ_2 retained close agreement between empirical SD and mean SE and 0.950 coverage. The principal finite-sample difficulty under heavy censoring is uncertainty estimation for the weaker treatment-specific dependence contrast rather than systematic bias across the full model. Full parameter-level results for all 12 settings are reported in Appendix A.

Table 3: Parameter-level Monte Carlo performance under moderate heterogeneity, $n = 250$, and 40% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	0.000	0.061	0.061	0.942
a_0	-0.005	0.152	0.152	0.940
a_1	0.012	0.209	0.202	0.937
a_2	0.007	0.210	0.195	0.923
σ_1	-0.024	0.099	0.093	0.899
b_0	-0.002	0.145	0.147	0.944
b_1	0.014	0.266	0.220	0.930
b_2	0.002	0.212	0.197	0.928
σ_2	-0.023	0.100	0.092	0.890
γ_0	-0.034	0.300	0.295	0.946
γ_1	0.091	0.638	0.527	0.923
γ_2	0.058	0.400	0.405	0.950

3.2 Constant-Copula Comparison

To assess the consequences of imposing homogeneous dependence, the same data-generating mechanism was considered at $n = 500$ and $n = 1000$ with 20% target censoring, while the fitted model was restricted to a common Clayton parameter,

$$\phi_i = \exp(\gamma_0). \tag{18}$$

The three association settings from Subsection 3.1 were retained. Under homogeneous dependence, the restriction is correctly specified; under moderate and

strong heterogeneity, the fitted common association represents a compromise across treatment groups. Full parameter-level results for both sample sizes are reported in Appendix B. Table 4 retains only the largest absolute bias among the four marginal treatment coefficients (a_1, a_2, b_1, b_2) and the minimum 95% coverage among the cure and marginal parameters.

Table 4: Key consequences of fitting a constant-dependence copula with 20% target censoring.

Association setting	n	Max |Bias| for treatment effects	Min 95% coverage
Homogeneous	500	0.008	0.933
	1000	0.009	0.935
Moderate heterogeneity	500	0.250	0.365
	1000	0.250	0.113
Strong heterogeneity	500	0.429	0.021
	1000	0.433	0.000

Under homogeneous dependence, the constant-copula fit maintained small bias and approximately nominal coverage at both sample sizes. In contrast, under moderate and strong heterogeneity, the maximum marginal treatment-effect biases remained nearly unchanged as n increased, while the minimum coverage deteriorated substantially. The largest distortions occurred for the Treatment 2 AFT coefficients in both margins; the susceptibility and scale estimates were comparatively stable in these simulation settings.

This pattern has a straightforward misspecified-likelihood interpretation. Although the susceptible marginal distributions are correctly specified, their parameters enter the copula contributions through F_1 and F_2. When heterogeneous dependence is incorrectly restricted to a common association parameter, joint likelihood estimation may therefore partially compensate by shifting the marginal coefficients. In large samples, the restricted estimator generally approaches the pseudo-true value

$$\boldsymbol{\Psi}_R^* = \arg\max_{\boldsymbol{\Psi}_R} E_0\{\ell_R(O_i; \boldsymbol{\Psi}_R)\}\,, \tag{19}$$

where E_0 denotes expectation under the true heterogeneous-dependence model and ℓ_R is the restricted log-likelihood contribution (White, 1982). The marginal components of $\boldsymbol{\Psi}_R^*$ need not equal their true values. Accordingly, larger samples can reduce estimation variability without removing the bias, and confidence intervals can become more concentrated around incorrect limiting values. Sandwich standard errors account for sampling variability around the fitted model's limiting parameters but do not eliminate this systematic misspecification bias. The persistence of bias and decline in coverage in Table 4 are consistent with this interpretation; the simulation comparison itself does not constitute a proof of the limiting behavior.

4 Data Analysis

We illustrate the model using the colon cancer trial data in the R `survival` package, which compared Observation, levamisole (Lev), and levamisole plus fluorouracil (Lev+5FU) (Laurie et al., 1989; Moertel et al., 1990). We define T_1 as time from study registration to recurrence and T_2 as time from recurrence to death. The analysis uses complete cases and is restricted to subjects compatible with this sequential recurrence-to-death ordering.

Both susceptible gap-time margins are modeled by log-logistic AFT distributions using the same treatment and clinical covariates. The susceptible proportion p is common to the sample, while residual association is modeled by

$$\phi_i = \exp\{\gamma_0 + \gamma_1 I(\text{Lev}_i) + \gamma_2 I(\text{Lev} + 5\text{FU}_i)\}. \tag{20}$$

The complete-case analysis contains 851 subjects. The estimated susceptible proportion was $\widehat{p} = 0.614$ (SE 0.040), corresponding to an estimated cure fraction of 0.386.

Table 5 reports the treatment coefficients from the two marginal AFT models. Lev showed little evidence of an association with either gap time. Lev+5FU was associated with a longer recurrence-free interval but a shorter post-recurrence survival interval. Full marginal-model estimates are given in Appendix C.

Table 5: Treatment effects in the two marginal AFT models.

	T_1: Registration to recurrence			T_2: Recurrence to death		
Treatment	Estimate	SE	95% CI	Estimate	SE	95% CI
Lev	-0.002	0.131	(−0.259, 0.255)	-0.150	0.125	(−0.395, 0.095)
Lev+5FU	0.410	0.180	(0.056, 0.763)	-0.380	0.150	(−0.674, −0.087)

Table 6 summarizes the treatment contrasts in the copula regression. Neither contrast was statistically significant, providing little evidence that treatment modifies the residual association between the two sequential gap times.

Table 6: Treatment contrasts for the conditional dependence regression.

Contrast	Estimate	SE	p-value	95% CI
Lev vs Observation	-0.151	0.738	0.838	(−1.598, 1.295)
Lev+5FU vs Observation	0.484	0.668	0.469	(−0.826, 1.793)

The corresponding Kendall's tau estimates were 0.109, 0.095, and 0.166 for Observation, Lev, and Lev+5FU, respectively, indicating weak positive dependence in all three groups. Overall, the application suggests that treatment may act differently across the two disease stages without materially changing their residual association.

5 Discussion

We proposed a fully parametric likelihood for two sequential gap times with a cure fraction and covariate-dependent Clayton association. The main simulation study showed small bias and generally satisfactory sandwich inference over the considered sample sizes, censoring levels, and associations. The constant-copula comparison adds an important qualification: when dependence truly differs across treatment groups, imposing homogeneous association can distort the marginal treatment coefficients despite correct specification of the susceptible marginal families. The additional $n = 1000$ experiment showed that the principal treatment-effect biases persisted as sample size increased, while confidence-interval coverage deteriorated. This behavior is consistent with convergence of the restricted estimator to pseudo-true, rather than necessarily true, marginal parameters. Thus, modeling association heterogeneity is relevant not only for describing dependence but also for the reliability of marginal treatment-effect inference.

In the colon cancer application, the two gap-time margins showed different treatment patterns, whereas the estimated residual dependence was weak and did not differ significantly across treatment groups. The strong consequences of misspecification in the simulation settings should therefore not be taken as evidence that comparable dependence heterogeneity exists in these data.

Several limitations remain. The susceptible probability p is common across treatment groups, and the Clayton family restricts attention to positive dependence with lower-tail emphasis. The application is also restricted to subjects compatible with the sequential recurrence-to-death ordering; deaths without a preceding recorded recurrence are not represented by this model and raise a separate semi-competing-risks issue (Fine et al., 2001). The misspecification experiment concerns a constant-association restriction within the Clayton family, not misspecification of the copula family or the marginal survival models. The observed persistence of bias offers a useful practical warning, but the exact pseudo-true parameters and their analytical bias were not derived here.

Overall, the proposed model extends sequential cure modeling by allowing residual association to vary with treatment or other covariates while retaining joint likelihood estimation. Its finite-sample performance and the focused constant-dependence comparison demonstrate why that flexibility can matter for treatment-effect estimation and its associated uncertainty assessment.

A Full Monte Carlo Results

Table 7: Monte Carlo results: Homogeneous dependence, $n = 250$, 20% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	-0.002	0.041	0.041	0.940
a_0	0.002	0.133	0.133	0.953
a_1	0.012	0.197	0.193	0.935
a_2	0.007	0.188	0.188	0.948
σ_1	-0.017	0.078	0.076	0.915
b_0	0.004	0.125	0.128	0.946
b_1	0.008	0.200	0.197	0.938
b_2	0.005	0.192	0.188	0.946
σ_2	-0.016	0.071	0.071	0.940
γ_0	-0.004	0.274	0.273	0.944
γ_1	-0.017	0.428	0.443	0.968
γ_2	0.001	0.425	0.422	0.956

Table 8: Monte Carlo results: Homogeneous dependence, $n = 500$, 20% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	-0.002	0.028	0.029	0.939
a_0	-0.001	0.092	0.093	0.950
a_1	0.002	0.143	0.136	0.929
a_2	-0.002	0.129	0.132	0.962
σ_1	-0.011	0.052	0.054	0.941
b_0	0.000	0.088	0.090	0.948
b_1	-0.003	0.142	0.139	0.947
b_2	0.002	0.135	0.133	0.940
σ_2	-0.010	0.051	0.050	0.940
γ_0	-0.017	0.188	0.191	0.965
γ_1	0.025	0.304	0.302	0.955
γ_2	0.017	0.287	0.286	0.957

Table 9: Monte Carlo results: Homogeneous dependence, $n = 250$, 40% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	-0.002	0.065	0.062	0.935
a_0	-0.005	0.152	0.155	0.951
a_1	0.006	0.205	0.204	0.943
a_2	0.012	0.196	0.196	0.951
σ_1	-0.023	0.102	0.098	0.913
b_0	0.003	0.150	0.148	0.953
b_1	0.010	0.239	0.228	0.933
b_2	0.007	0.203	0.202	0.937
σ_2	-0.016	0.101	0.095	0.915
γ_0	-0.004	0.303	0.293	0.940
γ_1	-0.020	1.229	0.577	0.927
γ_2	-0.007	0.589	0.470	0.958

Table 10: Monte Carlo results: Homogeneous dependence, $n = 500$, 40% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	0.000	0.044	0.044	0.941
a_0	0.001	0.108	0.109	0.953
a_1	0.008	0.151	0.144	0.934
a_2	0.003	0.140	0.139	0.957
σ_1	-0.010	0.073	0.070	0.923
b_0	0.003	0.103	0.104	0.949
b_1	0.014	0.173	0.162	0.927
b_2	0.005	0.148	0.142	0.938
σ_2	-0.008	0.070	0.069	0.937
γ_0	-0.005	0.203	0.204	0.957
γ_1	0.021	0.361	0.377	0.958
γ_2	0.015	0.309	0.311	0.955

Table 11: Monte Carlo results: Moderate heterogeneity, $n = 250$, 20% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	-0.004	0.042	0.041	0.931
a_0	0.001	0.131	0.131	0.958
a_1	0.006	0.198	0.193	0.933
a_2	-0.006	0.186	0.187	0.953
σ_1	-0.019	0.076	0.073	0.913
b_0	0.002	0.130	0.129	0.942
b_1	0.005	0.203	0.197	0.938
b_2	-0.003	0.191	0.188	0.942
σ_2	-0.015	0.074	0.070	0.915
γ_0	-0.017	0.293	0.274	0.935
γ_1	0.011	0.437	0.424	0.952
γ_2	0.035	0.378	0.366	0.955

Table 12: Monte Carlo results: Moderate heterogeneity, $n = 500$, 20% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	-0.001	0.029	0.029	0.948
a_0	0.001	0.092	0.092	0.943
a_1	-0.002	0.139	0.136	0.936
a_2	0.001	0.129	0.133	0.954
σ_1	-0.010	0.054	0.052	0.924
b_0	0.001	0.086	0.090	0.959
b_1	0.001	0.137	0.138	0.948
b_2	0.001	0.128	0.133	0.959
σ_2	-0.009	0.052	0.050	0.933
γ_0	-0.010	0.191	0.190	0.955
γ_1	0.014	0.288	0.287	0.953
γ_2	0.011	0.252	0.253	0.953

Table 13: Monte Carlo results: Moderate heterogeneity, $n = 250$, 40% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	0.000	0.061	0.061	0.942
a_0	-0.005	0.152	0.152	0.940
a_1	0.012	0.209	0.202	0.937
a_2	0.007	0.210	0.195	0.923
σ_1	-0.024	0.099	0.093	0.899
b_0	-0.002	0.145	0.147	0.944
b_1	0.014	0.266	0.220	0.930
b_2	0.002	0.212	0.197	0.928
σ_2	-0.023	0.100	0.092	0.890
γ_0	-0.034	0.300	0.295	0.946
γ_1	0.091	0.638	0.527	0.923
γ_2	0.058	0.400	0.405	0.950

Table 14: Monte Carlo results: Moderate heterogeneity, $n = 500$, 40% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	-0.001	0.044	0.043	0.948
a_0	-0.008	0.109	0.106	0.941
a_1	0.009	0.147	0.145	0.959
a_2	0.011	0.140	0.139	0.948
σ_1	-0.013	0.071	0.066	0.918
b_0	-0.009	0.106	0.103	0.945
b_1	0.018	0.163	0.156	0.940
b_2	0.013	0.143	0.140	0.933
σ_2	-0.013	0.071	0.066	0.916
γ_0	0.001	0.199	0.201	0.949
γ_1	0.049	0.356	0.353	0.945
γ_2	0.004	0.275	0.275	0.945

Table 15: Monte Carlo results: Strong heterogeneity, $n = 250$, 20% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	-0.003	0.042	0.040	0.942
a_0	-0.003	0.134	0.129	0.944
a_1	0.003	0.195	0.192	0.937
a_2	-0.008	0.192	0.187	0.936
σ_1	-0.022	0.076	0.069	0.900
b_0	0.002	0.130	0.128	0.948
b_1	-0.004	0.195	0.193	0.952
b_2	-0.012	0.191	0.187	0.930
σ_2	-0.022	0.074	0.068	0.883
γ_0	-0.018	0.275	0.273	0.956
γ_1	0.023	0.389	0.391	0.955
γ_2	0.043	0.366	0.357	0.952

Table 16: Monte Carlo results: Strong heterogeneity, $n = 500$, 20% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	-0.002	0.029	0.028	0.932
a_0	-0.000	0.092	0.091	0.952
a_1	-0.001	0.140	0.137	0.941
a_2	-0.004	0.140	0.132	0.939
σ_1	-0.013	0.052	0.050	0.920
b_0	-0.003	0.092	0.090	0.945
b_1	0.004	0.141	0.138	0.943
b_2	-0.001	0.141	0.133	0.930
σ_2	-0.012	0.052	0.049	0.907
γ_0	-0.002	0.188	0.189	0.952
γ_1	0.004	0.277	0.268	0.951
γ_2	0.012	0.243	0.244	0.952

Table 17: Monte Carlo results: Strong heterogeneity, $n = 250$, 40% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	-0.004	0.059	0.060	0.947
a_0	-0.010	0.147	0.149	0.944
a_1	0.004	0.217	0.204	0.934
a_2	0.012	0.199	0.196	0.953
σ_1	-0.027	0.095	0.089	0.892
b_0	-0.006	0.144	0.146	0.939
b_1	0.007	0.228	0.215	0.938
b_2	0.011	0.198	0.198	0.948
σ_2	-0.024	0.096	0.090	0.892
γ_0	-0.018	0.301	0.296	0.955
γ_1	0.106	0.507	0.493	0.935
γ_2	0.059	0.396	0.399	0.950

Table 18: Monte Carlo results: Strong heterogeneity, $n = 500$, 40% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	-0.003	0.042	0.042	0.942
a_0	-0.006	0.100	0.104	0.954
a_1	-0.003	0.147	0.145	0.954
a_2	0.001	0.135	0.138	0.963
σ_1	-0.016	0.064	0.063	0.927
b_0	-0.002	0.100	0.102	0.949
b_1	-0.007	0.154	0.151	0.948
b_2	-0.002	0.136	0.139	0.956
σ_2	-0.014	0.066	0.064	0.930
γ_0	-0.020	0.211	0.204	0.946
γ_1	0.060	0.348	0.332	0.925
γ_2	0.024	0.271	0.269	0.947

B Constant-Copula Comparison Results

Table 19: Constant-copula results: Homogeneous dependence, $n = 500$, 20% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	-0.002	0.028	0.029	0.943
a_0	-0.005	0.087	0.087	0.948
a_1	0.008	0.122	0.117	0.949
a_2	0.003	0.112	0.112	0.945
σ_1	-0.010	0.052	0.054	0.943
b_0	-0.004	0.082	0.084	0.947
b_1	0.004	0.127	0.122	0.948
b_2	0.007	0.119	0.114	0.933
σ_2	-0.009	0.051	0.050	0.940

Table 20: Constant-copula results: Moderate heterogeneity, $n = 500$, 20% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	0.002	0.029	0.029	0.947
a_0	-0.096	0.088	0.087	0.796
a_1	0.071	0.122	0.117	0.902
a_2	0.250	0.105	0.107	0.365
σ_1	-0.003	0.056	0.053	0.940
b_0	-0.096	0.085	0.085	0.785
b_1	0.074	0.126	0.121	0.901
b_2	0.244	0.108	0.109	0.399
σ_2	-0.004	0.053	0.051	0.936

Table 21: Constant-copula results: Strong heterogeneity, $n = 500$, 20% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	0.005	0.030	0.029	0.932
a_0	-0.195	0.088	0.087	0.395
a_1	0.218	0.122	0.115	0.535
a_2	0.429	0.115	0.107	0.021
σ_1	0.006	0.054	0.052	0.936
b_0	-0.198	0.086	0.086	0.371
b_1	0.219	0.124	0.118	0.556
b_2	0.425	0.117	0.107	0.026
σ_2	0.002	0.054	0.051	0.931

Table 22: Constant-copula results: Homogeneous dependence, $n = 1000$, 20% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	-0.001	0.021	0.020	0.942
a_0	-0.005	0.063	0.062	0.945
a_1	0.003	0.086	0.083	0.938
a_2	0.009	0.082	0.080	0.951
σ_1	-0.005	0.039	0.039	0.936
b_0	-0.004	0.060	0.059	0.944
b_1	0.002	0.088	0.087	0.950
b_2	0.006	0.084	0.081	0.947
σ_2	-0.006	0.036	0.036	0.935
γ_0	-0.001	0.093	0.092	0.951

Table 23: Constant-copula results: Moderate heterogeneity, $n = 1000$, 20% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	0.001	0.020	0.020	0.959
a_0	-0.096	0.064	0.061	0.644
a_1	0.073	0.088	0.083	0.845
a_2	0.250	0.080	0.077	0.113
σ_1	0.002	0.039	0.038	0.938
b_0	-0.097	0.061	0.060	0.614
b_1	0.072	0.088	0.086	0.877
b_2	0.245	0.079	0.077	0.122
σ_2	0.000	0.038	0.036	0.943

Table 24: Constant-copula results: Strong heterogeneity, $n = 1000$, 20% target censoring.

Parameter	Bias	Empirical SD	Mean SE	95% Coverage
p	0.006	0.019	0.020	0.943
a_0	-0.196	0.060	0.061	0.111
a_1	0.221	0.081	0.082	0.237
a_2	0.433	0.078	0.076	0.000
σ_1	0.009	0.037	0.037	0.945
b_0	-0.197	0.060	0.061	0.099
b_1	0.218	0.085	0.084	0.252
b_2	0.428	0.079	0.076	0.000
σ_2	0.006	0.037	0.036	0.941

C Full Marginal Estimates for the Colon Cancer Analysis

Table 25: Adjusted estimates from the two log-logistic AFT marginal models.

	T_1: Registration to recurrence			T_2: Recurrence to death		
Covariate	Estimate	SE	95% CI	Estimate	SE	95% CI
Intercept	8.764	0.760	(7.275, 10.253)	8.991	0.547	(7.919, 10.062)
Lev	-0.002	0.131	(−0.259, 0.255)	-0.150	0.125	(−0.395, 0.095)
Lev+5FU	0.410	0.180	(0.056, 0.763)	-0.380	0.150	(−0.674, −0.087)
Sex	0.270	0.123	(0.029, 0.512)	-0.144	0.110	(−0.359, 0.071)
Age	-0.001	0.004	(−0.009, 0.008)	-0.018	0.005	(−0.028, −0.009)
Obstruction	-0.426	0.152	(−0.724, −0.129)	-0.280	0.137	(−0.549, −0.011)
Perforation	-0.057	0.205	(−0.459, 0.345)	0.382	0.294	(−0.195, 0.958)
Adherence	-0.045	0.144	(−0.328, 0.238)	-0.062	0.134	(−0.325, 0.200)
Nodes	-0.017	0.022	(−0.061, 0.027)	-0.030	0.020	(−0.069, 0.010)
Differentiation	-0.299	0.120	(−0.534, −0.063)	-0.169	0.113	(−0.390, 0.053)
Extent	-0.582	0.199	(−0.971, −0.193)	-0.344	0.134	(−0.607, −0.082)
Surgery timing	-0.105	0.137	(−0.374, 0.164)	-0.071	0.114	(−0.293, 0.152)
Node4	-0.587	0.202	(−0.984, −0.190)	-0.447	0.159	(−0.758, −0.136)
Scale	0.630	0.054	(0.525, 0.735)	0.636	0.030	(0.577, 0.695)

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